MATHS MANTHAN
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 06
Chapter 10: Circles

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. In a circle of radius 7 cm, tangent PT is drawn from external point P. If $OT = 7$ cm and $PT = 24$ cm, then the distance OP is: [1]

(A) 25 cm (B) 26 cm (C) 31 cm (D) $\sqrt{527}$ cm

Q2. If two tangents inclined at an angle of $60°$ are drawn to a circle of radius 3 cm, then the length of each tangent is: [1]

(A) $\frac{3}{2}\sqrt{3}$ cm (B) 6 cm (C) 3 cm (D) $3\sqrt{3}$ cm

Q3. If a parallelogram circumscribes a circle, then the parallelogram must be a: [1]

(A) Rectangle (B) Square (C) Rhombus (D) Trapezium

Q4. Two concentric circles have radii a and b ($a > b$). A chord of length $2c$ of the larger circle touches the smaller circle. The relation between a, b, c is: [1]

(A) $a^2 = b^2 + c^2$ (B) $b^2 = a^2 + c^2$ (C) $c^2 = a^2 + b^2$ (D) $a + b = 2c$

Q5. **Assertion (A):** At most two tangents can be drawn to a circle from a point lying outside the circle.
Reason (R): From a point lying strictly inside the circle, exactly one tangent can be drawn to the circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Prove that the tangents drawn at the end points of a diameter of a circle are parallel. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre. **[3]**

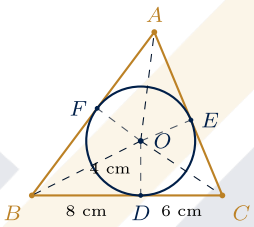
Q8. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. Find the length PQ and the area of $\triangle OPQ$. **[3]**

Q9. In a quadrilateral $ABCD$ circumscribing a circle, prove that $\angle AOB + \angle COD = 180°$ and $\angle BOC + \angle AOD = 180°$, where O is the centre of the circle. **[3]**

SECTION D: Case-Based / Applied Competency Question (1 × 4 = 4 Marks)

Q10. Municipal Triangular Park with Inscribed Circular Fountain

A municipal corporation plans an ornamental circular fountain inscribed within a triangular park ABC. The park boundaries measure $AB = 15$ m, $BC = 14$ m, $AC = 13$ m. The fountain touches boundary BC at point D, AC at E, and AB at F.



(i) Calculate the semi-perimeter s of the triangular park. **[1]**

(ii) Find the lengths of tangent segments BD and CD on side BC. **[1]**

(iii) Calculate the radius r of the circular fountain if the area of $\triangle ABC = 84$ m^2, OR find the length of tangent segment AF. **[2]**

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